NUMERICAL SIMULATIONS OF THE RADIO-FREQUENCY–DRIVEN TOROIDAL CURRENT IN TOKAMAKS

Y. PEYSSON* and J. DECKER

CEA, IRFM, F-13108, Saint-Paul-lez-Durance, France

Received March 7, 2013
Accepted for Publication June 14, 2013
doi:10.13182/FST13-643

Radio-frequency (rf) waves are a powerful tool for improving the performance and stability of tokamak plasmas through heating and current drive mechanisms, allowing current density profile control and steady-state operation. From first principles, and taking advantage from the ordering between the various time and space scales, fast and powerful numerical tools have been developed to calculate the rf-driven current. The current drive problem in tokamaks is first introduced with the purpose of maintaining a steady-state self-organized toroidal magnetohydrodynamic equilibrium, such that a minimal amount of the fusion power has to be recycled to control the plasma current. The strict criterion that characterizes a steady-state discharge is derived from the response of the tokamak, considered as a transformer, and of the plasma, when an external source of current is applied. The calculation of a rf-driven source of current requires solving self-consistently a set of equations describing the dynamics of wave fields and charged particles in an inhomogeneous magnetized plasma. The range of applicability of these equations is discussed, as well as numerical methods developed to solve them, such as the ray-tracing code C3PO and the three-dimensional linearized relativistic bounce-averaged electron Fokker-Planck solver LUKE. Simulations of current drive by lower-hybrid waves are presented to illustrate the applications of our numerical tools. Current drive modeling includes the effect of electron density fluctuations at the plasma edge, and the case of electron cyclotron waves used for stabilization of the 3/2 neoclassical tearing modes in ITER is studied in detail. Finally, ongoing developments, including cross effects between momentum and configuration spaces, aiming at improving current drive calculations are discussed.

KEYWORDS: *electromagnetic wave propagation, theory and numerical simulation, rf current drive*

Note: Some figures in this lecture may be in color only in the electronic version.

I. INTRODUCTION

The toroidal plasma current is a key parameter of the tokamak physics. Its noninductive generation allows steady-state operation, while the control of its radial profile leads to improved fusion performances by modifying locally the winding of the magnetic field lines in the plasma.

Among various possible methods to generate a toroidal current, those based on the use of radio-frequency (rf) waves are certainly the most powerful and flexible used so far, combining a high current drive efficiency (CD) in amperes per watt with control over the location and spread of power deposition. Moreover, as the rf power sources may be placed far from the plasma, rf waves are particularly convenient in the complex environment of a fusion reactor such as ITER (Ref. 1).

Most attractive rf-driven current methods result from the resonant interactions between fast (weakly collisional) electrons in the plasma and the oscillatory electric field of

*E-mail: yves.peysson@cea.fr

propagating high-frequency rf waves.[2] Waves with frequencies $\omega/2\pi \lesssim 10$ GHz, equal to or lower than the lower-hybrid (LH) frequency, transfer momentum to electrons in the direction parallel to the magnetic field through the Landau resonance mechanism. The use of rf waves for heating or driving a current in plasmas was proposed by Stix.[3] The first experimental evidences were observed ~15 years later on the small tokamak Versator II, where it has been possible to reverse the loop voltage and demonstrate that LH waves could be efficient enough in driving a current in the plasma and recharge the primary of the transformer.[4] Conversely, higher-frequency waves in the electron cyclotron (EC) frequency range $\omega/2\pi \sim 100$ GHz transfer mostly perpendicular momentum to electrons, yet driving a parallel current resulting from the rf-induced asymmetric resistivity. Pioneering theoretical work from Fisch and Boozer[5] shows that a direct source of parallel momentum is not necessary for efficient current generation, as demonstrated experimentally soon after.[6] Electron cyclotron waves are also an essential tool for controlling magnetohydrodynamic (MHD) modes in the ITER tokamak.[7]

The current source is calculated by solving self-consistently the wave fields (Maxwell equations) and the electron velocity distribution (Fokker-Planck equation).[8] Owing to the complexity of the problem characterized by multiple time and space scales, the rf-driven current and the corresponding power deposition are calculated by numerical codes. The development of these tools is made a challenging task by the complexity of the underlying equations, the necessary use of curvilinear metrics (for noncircular shaped plasmas), and the existence of synergistic effects between multiple current sources. The calculations increasingly rely on parallelized computing to reduce calculation time as rf-driven current codes are elements of a chain of codes used in integrated time-dependent simulations of the plasma. Several suites of codes addressing integrated modeling of rf heating and current drive in tokamak plasmas have been developed, including METIS/CRONOS (Ref. 9), LSC (Ref. 10), ACCOME (Ref. 11), GENRAY/CQL3D (Refs. 12 and 13), or TASK (Ref. 14). In METIS/CRONOS, the rf-driven currents are principally calculated by the ray-tracing code C3PO coupled to the three-dimensional (3-D) linearized relativistic bounce-averaged electron Fokker-Planck solver LUKE (Refs. 15 and 16). The present lecture is extensively based on the theoretical and numerical developments related to C3PO/LUKE, even though most of the concepts discussed in this lecture are valid beyond these two codes.

This lecture is divided into six sections. In Sec. II, general aspects of current generation in tokamak plasmas are first presented, with the purpose of maintaining a steady-state self-organized toroidal MHD equilibrium, such that a minimal amount of the fusion power has to be recycled to control the plasma current. The strict criterion that characterizes a steady-state discharge is derived from the response of the tokamak, considered as a transformer, and of the plasma, when an external source of current is applied. By an appropriate ordering between the different time and space scales, the equations used to calculate the rf current source are introduced in Sec. III. The corresponding limitations are discussed. The rf current source module based on the C3PO ray-tracing code and the 3-D linearized relativistic bounce-averaged electron Fokker-Planck solver LUKE corresponds to rigorous numerical implementations of these equations. These codes, presented in Sec. IV, have been extensively benchmarked and are used routinely for tokamak simulations. Some prominent numerical LH current drive calculations are detailed in Sec. V. The highly modular current source module described in Sec. IV includes the effect of density fluctuations at the plasma edge. This point is discussed in Sec. VI, where the stabilization of 3/2 neoclassical tearing modes (NTMs) in ITER by EC waves is studied. Before concluding, recent theoretical developments carried out for improving current drive calculations by including cross effects between momentum and configuration spaces are briefly discussed in Sec. VII.

II. CURRENT DRIVE PROBLEM IN TOKAMAKS

II.A. Need for a Self-Organized Steady-State Tokamak

The toroidal plasma current I_p plays a central role in the physics of tokamaks. It allows the existence of a toroidal MHD equilibrium through the radial force balance equation $\mathbf{j}_p \times \mathbf{B} = \nabla p$, where $\mathbf{j}_p$ is the local current density flowing in the plasma,[a] $\mathbf{B}$ is the magnetic field, and p is the plasma pressure. As the energy confinement time τ_E scales roughly as a power of I_p (Ref. 17), maintaining a high plasma current is also crucial to satisfying the Lawson criterion and obtaining a positive fusion power balance toward an economically viable energy. The underlying mechanism for performance and stability is the winding of the magnetic field lines in the plasma, which is characterized by the safety factor $q(\psi) \equiv d\phi/d\theta$, where ϕ is the toroidal angle and θ the poloidal angle. An appropriate control of the $q(\psi)$ profile, which is primarily determined by $\mathbf{j}_p$, can improve the stability and performance of the plasma significantly by reducing the turbulence, leading to promising operating regimes for a future fusion-based electrical power plant. In this context, $\mathbf{j}_p$ is a critical parameter, and its accurate and continuous control is an essential part of tokamak operation.

[a] $I_p = \int\int \mathbf{j}_p \cdot \hat{\phi} dS$, where dS is the elementary surface perpendicular to the toroidal direction $\hat{\phi}$.

Based on the electrical transformer concept, the original tokamak operating mode is intrinsically transient, which represents a major drawback for economic reasons, but also imposes technical limitations, as cyclic operation can lead to excessive constraints on the mechanical structure. This essential difficulty is overcome by driving $\mathbf{j}_p$ continuously. This can be achieved in two ways as the plasma current $\mathbf{j}_p = \mathbf{j}_{conf} + \mathbf{j}_{ext}$ is the sum of a configuration current $\mathbf{j}_{conf}$, arising to prevent electrical charge accumulation in the plasma as $\nabla \cdot \mathbf{j}_{conf} = 0$, and a current $\mathbf{j}_{ext}$ externally imposed by rf waves, beam injection, etc.[b] Since $\nabla \cdot \mathbf{j}_{conf\perp} \neq 0$, where $\mathbf{j}_{conf\perp} = \mathbf{j}_{conf} \times \hat{b}$ is the diamagnetic current flowing perpendicularly to the field line $\hat{b} = \mathbf{B}/\|\mathbf{B}\|$, the existence of a plasma pressure gradient ∇p implies that a parallel current $\mathbf{j}_{conf\parallel} = (\mathbf{j}_{conf} \cdot \hat{b})\hat{b}$ must be flowing along the magnetic field lines. At low plasma collisionality, $\mathbf{j}_{conf\parallel} \sim \nabla p$ can be identified to the usual bootstrap current[18–20] resulting from pressure anisotropy.[c]

In the early developments of tokamak scenarios, steady-state operation was supposed to be carried preferentially by external means, neglecting the configuration current $\mathbf{j}_{conf}$.[d] In this case, the current drive efficiency, defined by the ratio $\eta = |\mathbf{j}_{ext}|/P_{ext}$, where P_{ext} is the external power associated with $\mathbf{j}_{ext}$, is a critical engineering parameter to qualify a current drive method, since τ_E scales approximately as $1/\sqrt{\Sigma P_{ext}}$ (Ref. 17), where ΣP_{ext} is the sum of all sources of external power injected in the plasma for heating and current drive. Indeed, poor η values may lead to a drastic reduction of the fusion power efficiency $Q = P_{fusion}/\Sigma P_{ext}$, if most of the fusion power P_{fusion} must be recycled to drive all the plasma current. Unfortunately, extensive theoretical studies validated by numerous experimental investigations have shown that none of the current drive methods have the efficiency required to reach the goal of a steady-state discharge sustained predominantly by external means without consuming most of the fusion power.[21] As an example, recent simulations have shown that 20 MW of LH power drives no more than 1 MA of current in the ITER tokamak, which represents only a small fraction I_p whatever the chosen scenario.[22]

Ideally, the contribution of $\mathbf{j}_{ext}$ to $\mathbf{j}_p$ must be kept to a minimum required for steady-state tokamak operation. Since $\mathbf{j}_{conf}$ must predominate over $\mathbf{j}_{ext}$, plasmas with large pressure gradients ∇p much be achieved, while the role of external means is restricted to maintain the self-consistency between the pressure and current profiles at all times for a stable toroidal MHD equilibrium. For these so-called advanced tokamak scenarios,[23,24] the total plasma current I_p is also lowered to reduce $\mathbf{j}_{ext}$ as much as possible, assuming that the corresponding reduction of τ_E can be compensated by the onset of an improved confinement related to internal transport barriers.[25,26] This self-organized steady-state regime has been extensively investigated experimentally.[24,27] While being obviously attractive, such highly nonlinear configurations are often only marginally stable,[28,29] which increases the role played by external means to control the toroidal plasma current profile and maintain continuously the consistency between $\mathbf{j}_p$ and ∇p.

With this new paradigm for steady-state tokamak operation, a high efficiency η becomes a less critical criterion to qualify a current drive method as compared to its capability to perform an accurate real-time control of the current density profile.[30–33] Consequently, the possibility to drive a current anywhere from the core to the edge of the plasma with either broad or narrow profiles by adjusting tunable parameters is a highly desirable asset. The properties of rf waves make them particularly attractive for this purpose.

Therefore, simulating self-organized steady-state tokamak operation puts very high constraints on the level of accuracy that should be reached by current source calculations, not only for the interpretation of the observed phenomenology but also to make reliable predictions. Consequently, the calculations of any current source cannot be performed outside the framework of integrated tokamak modeling, to satisfy the consistency between $\mathbf{j}_p$ and ∇p, which evolve at different timescales. Since a single type of rf wave cannot fulfill all requirements for the current generation, a unified multiwave description must be considered to tackle the difficult problem of synergy between the different methods. In addition, neoclassical corrections due to locally large ∇p must be also considered. Since codes that calculate the current sources are included in intricate integrated chains of codes,[34,35] numerical tools designed from advanced physics considerations must be fast, robust, and possibly modular, depending upon the separation between the various time and space scales. In addition, computations must take benefit of latest progress in computing hardware (parallel processing, graphics processing units), a considerable challenge in itself.

II.B. From Transient to Steady Tokamak Operation

The achievement of a steady-state regime is a complex process, which involves the response of the tokamak as a transformer and that of the plasma to an external source of current. This problem has been studied in detail in the Tore Supra tokamak, ideally designed for long-pulse operation with a toroidal magnetic field

[b] By assuming the additivity of $\mathbf{j}_{conf}$ and $\mathbf{j}_{ext}$, it is supposed that possible synergistic effects between the two contributions are negligible. This assumption is discussed in Sec. VII.

[c] At the plasma edge, the temperature is low enough to make the collisional operator almost isotropic. In that case, $\mathbf{j}_{conf\parallel}$ represents the Pfirsch-Schlüter current. The corresponding flux-surface–averaged toroidal current is usually negligible, except if the poloidal cross section of the plasma is not circular.

[d] In those days, most of the tokamak plasmas were operated at low β values, such that the configuration current represented $<10\%$ of I_p. Here, β is the ratio of the kinetic to the magnetic pressure.

produced by superconducting coils.[36] At constant I_p, the plasma duration is limited by the consumption of the available magnetic flux Φ_M of the transformer, according to the simple law

$$\frac{d\Phi_M}{dt} = L_0 \frac{dI_0}{dt} = -\mathcal{R}_p \frac{L_0}{M}\left(I_p - I_{ext}^{ni}\right) , \qquad (1)$$

where

L_0 = magnetic inductance of the primary circuit

M = mutual inductance between the primary circuit and the plasma, which plays the role of a secondary circuit

$\mathcal{R}_p$ = plasma resistance.

The equivalent circuit is shown schematically in Fig. 1. The duration of the discharge is limited by the upper bound of the current I_0 in the primary circuit. By heating the plasma, its duration may be significantly increased as $\mathcal{R}_p \sim T_e^{-3/2}$, where T_e is the volume-averaged electron temperature. Yet, a fully steady-state regime implies that $dI_0/dt = 0$ exactly, such that $I_p = I_{ext}^{ni}$, where I_{ext}^{ni} is a noninductive source of current. Since η is never exactly constant during a discharge, a fixed P_{ext} does not guarantee that $I_p = I_{ext}^{ni}$—the primary criterion for full current drive operation—may be achieved rigorously. Analysis of the transformer circuit equation eigenmodes shows that the condition $I_p = I_{ext}^{ni}$ can be rapidly achieved if Φ_M is feedback controlled to a constant value, as the transition between the inductive and the noninductive regimes takes place on a relatively short timescale $\tau_T = L_p\left[1 - M^2/\left(L_0 L_p\right)\right]/\mathcal{R}_p$, where L_p is the magnetic inductance of the plasma. For the Tore Supra tokamak, $\tau_T \simeq 1.3$ s, in good agreement with experimental observations, as shown in Fig. 2. In this operating mode, the level of I_p is free but can be adjusted to the required level by an appropriate feedback control of P_{ext} depending upon the η value, while the loop voltage is exactly 0 V.

Nevertheless, the condition $dI_0/dt = 0$ does not imply that a steady state is effectively achieved in the plasma regarding the current density profile. In the large aspect ratio limit and for circular concentric magnetic flux surfaces, it can be shown that the space-time evolution of the toroidal electric field E_ϕ is given by the following one-dimensional equation, resulting from a combination of the Ampere, Faraday, and Ohm laws:

$$\frac{1}{\rho}\frac{\partial}{\partial \rho}\left(\rho \frac{\partial E_\phi}{\partial \rho}\right) = \frac{\partial}{\partial \tau}\left(E_\phi + \frac{j_{ext,\phi}^{ni}}{\sigma_\phi}\right) , \qquad (2)$$

where

ρ = r/a = normalized minor radius; r ranges from 0 at the plasma center to a at the separatrix

τ = time normalized to the resistive time $\tau_r = \mu_0 \sigma_\phi a^2$

σ_ϕ = toroidal electrical conductivity.

From Eq. (2), the steady-state regime corresponds to the unique definition $E_\phi = 0$, everywhere in the plasma, which is a higher constraint than a zero loop voltage measured at the plasma boundary.

When a source of current $j_{ext,\phi}^{ni}$ is added suddenly at some location, an electric field E_ϕ appears immediately to compensate exactly the local excess of current. The initial net effect of any external source of current on I_p is therefore null, whatever the efficiency η of the method. Then, the current density profile starts to evolve until a steady state is reached. The slowest spatial evolution corresponds to the fundamental eigenmode of Eq. (2), its timescale being $\tau_r^* = \tau_r/\lambda_1^2$, where $J_1(\lambda_1) = 0$ and $J_1(x)$ is the Bessel function of first order. For hot plasmas with a large cross section, τ_r^* may be very large owing to the $T_e^{3/2}$ dependence of σ_ϕ and significantly exceeds τ_T. The simple law $\sigma_\phi \simeq 6 \times 10^8 T_{e(\mathrm{keV})}^{3/2}/\ln\Lambda$ ($\Omega^{-1}\cdot\mathrm{m}^{-1}$) (Ref. 37),

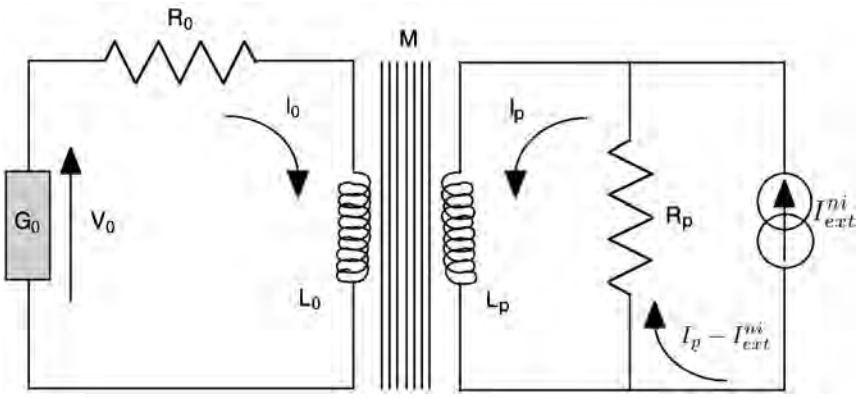


Fig. 1. Schematic view of the tokamak considered as an electrical transformer. The transformer is schematically represented by two circuits, which gives reasonable quantitative agreement with experimental observations.[36] (a) The primary circuit and (b) the plasma ring, which plays the role of the secondary circuit.

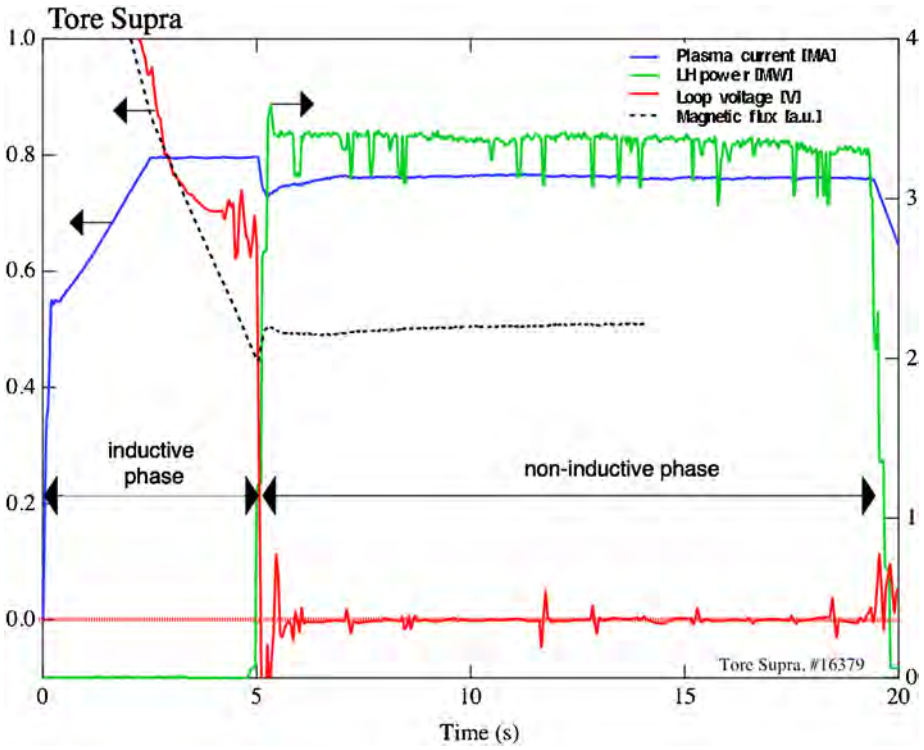


Fig. 2. Time evolution of the plasma current I_p, the LH power P_{LH}, the loop voltage V_{loop}, and the magnetic flux ϕ_M for Tore Supra discharge #16379. Before $t = 5$ s during the inductive phase, the magnetic consumption is important and ϕ_M decreases rapidly. Once ϕ_M is feedback controlled to a constant value, the loop voltage drops to zero in <1.5 s, and the magnetic flux consumption is fully stopped. The plasma current is slightly reduced as the current drive efficiency does not maintain the initial current level with 3.5 MW of LH power. The duration of this type of discharge is only limited by the LH power source, and a 6-min discharge has been successfully achieved in the Tore Supra tokamak by this technique.[94]

where $\ln\Lambda$ is the coulomb logarithm, yields $\tau_r^* \approx 17$ s in Tore Supra for $T_e = 5$ keV and $a = 0.7$ m.

In ITER, τ_r^* may reach hundreds of seconds depending upon the mean plasma electron temperature and could represent more than one-third of the foreseen plasma duration in the flattop for the standard ELMy H-mode scenario characterized by edge-localized mode MHD activity.[22] In this context, if the current density profile has not a consistent shape with ∇p, it will evolve continuously and extremely slowly during the flap-top phase of the discharge, which represents a risk regarding the MHD stability and may alter fusion performances. To avoid or limit any large mismatch once the plasma is self-heated by alpha particles in the thermonuclear regime, a fast preshaping of the current density profile has likely to be considered in the initial phase of the ramp-up, when the temperature is low enough for the current density profile to evolve on a sufficiently short timescale.[38e] This represents a challenging issue, which can however be investigated in machines of smaller size because their plasma temperature is usually much lower. Therefore, current drive studies performed in existing tokamaks are fully pertinent regarding ITER operation, even though the respective steady-state regimes are ultimately completely different. Indeed, in existing machines, steady-state operation is predominantly maintained by external means, while in ITER, their contributions must be minimal.

II.C. Timescales and Current Drive Modeling

As the resistive diffusion described by Eq. (2) takes place on a very long timescale, the toroidal MHD equilibrium can be considered as being quasi-static with respect to the main physical processes taking place in the integrated tokamak modeling, which are heat and particle transport.[9] The equilibrium is also quasi-static regarding the evolution of rf current sources, since all corresponding characteristics times ($2\pi/\omega$, τ_o, and ν_{coll}^{-1}) are usually much shorter than τ_E, τ_T, and τ_r^*. Here, ω is the wave angular frequency, τ_o is the time for fast electrons to complete their closed orbits in a poloidal plane, and ν_{coll} is the electron-ion coulomb collision frequency.[8,39] For hot plasmas of large size, the ordering $2\pi/\omega \ll \tau_o \ll \nu_{coll}^{-1} \ll \tau_E \ll \tau_T \ll \tau_r^*$ usually holds over a wide cross

[e]A cyclic scenario has been also proposed as an alternative approach.[63]

section of the plasma, and considerable simplifications can be carried out in the models for a consistent description of a tokamak discharge in the presence of external rf current sources. The level of modularity that can be reached in the integrated modeling, shown in Fig. 3, and the possibility to link different numerical codes for accurate calculations, are therefore a direct consequence of the physical timescale separability.

III. EQUATIONS FOR CALCULATING A rf CURRENT SOURCE

The calculation of the rf-driven current is based on the self-consistent resolution of the Boltzmann equation for the particles (electrons and ions) and the Maxwell equations for the fields.[8,39]

III.A. The 3-D Linearized Bounce-Averaged Fokker-Planck Equation

Since $m_e/m_i \ll 1$, where $m_{e[i]}$ is the electron [ion] rest mass, ion dynamics is ignored. As weak deflections dominate coulomb interactions in tokamak plasmas, collisions between charged particles are described by a Fokker-Planck operator $\mathcal{C}$. The accuracy of the current drive calculations is essentially limited to $1/\ln\Lambda$ since large angle collisions are neglected. The current density at time t and spatial position $\mathbf{x}$ is $\mathbf{j}(\mathbf{x},t)=(-e/m_e)\iiint d^3\mathbf{p} f(\mathbf{x},\mathbf{p},t)\mathbf{p}/\gamma$, and the corresponding power deposition is $P(\mathbf{x},t)=m_e c^2 \iiint d^3\mathbf{p}/(\gamma-1)\partial f(\mathbf{x},\mathbf{p},t)/\partial t$. Both quantities are moments of the same single-particle electron distribution function $f(\mathbf{x},\mathbf{p},t)$, where $\mathbf{p}$ is the electron momentum and $\gamma=\sqrt{1+(p/mc)^2}$ is the relativistic factor, with $p=\|\mathbf{p}\|$. The function $f(\mathbf{x},\mathbf{p},t)$ is the solution of the Fokker-Planck equation,

$$\frac{df}{dt}=\frac{\partial f}{\partial t}+\dot{\mathbf{x}}\cdot\nabla_{\mathbf{x}}f+\dot{\mathbf{p}}\cdot\nabla_{\mathbf{p}}f=\mathcal{C}(f)\,,$$

where

$\dot{\mathbf{x}} = \mathbf{v} = \mathbf{p}/(\gamma m)$ is the electron velocity

$\dot{\mathbf{p}} = e(\mathbf{E}+\mathbf{v}\times\mathbf{B})$ is the Lorentz force,

and where the operator $\mathcal{C}$ conserves particles.

The resolution of this system of equations can be simplified by applying various orderings. While orderings restrict the range of applicability of the model, they reduce the dimensionality of the problem for a faster numerical resolution. The first ordering is defined by the small parameter $\delta^2\equiv\rho_L/a$, where ρ_L is the thermal Larmor radius and a is the tokamak minor radius. The ordering $\delta^2\ll 1$ is used to separate the electron gyromotion from the guiding-center dynamics. To order zero in δ, $\mathbf{j}_0(\mathbf{x},t)$, a moment of the gyroindependent distribution function $f_0(\mathbf{x},\mathbf{p},t)$, is the macroscopic current entering the toroidal MHD equilibrium equation. To first order in δ, the constitutive relation $\mathbf{j}_1(\mathbf{x},t)=\mathbb{S}(f_0)\mathbf{E}_1$ is obtained where the first-order perturbed electric field $\mathbf{E}_1$ is a solution to the linearized Maxwell equation system, where dielectric properties of the plasma are described by the conductivity tensor $\mathbb{S}(f_0)$. To second order in δ, the equation describing the slow time evolution of the guiding-center distribution f_0 is obtained by averaging over the fast timescales, using the linear relation between f_0 and f_1 at the first order. The

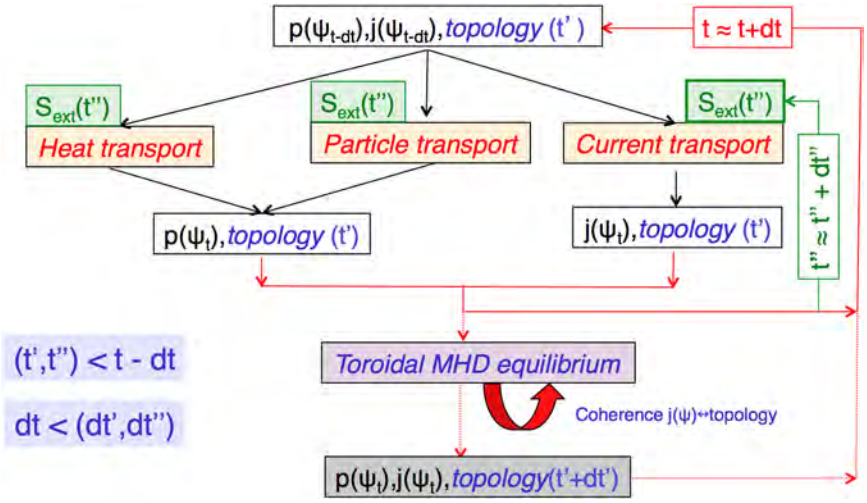


Fig. 3. Schematic code structure for integrated tokamak modeling.[9] From a given toroidal MHD equilibrium, the heat, particle, and current sources (including by rf waves) are calculated before solving the heat, particle, and current transport equations. Depending upon the different timescales and the evolution of the plasma shape (topology), some sources are recalculated before the determination of a new toroidal MHD equilibrium. It contributes to save computational effort. Because of the code interplay, the numerical robustness of each module is critical.

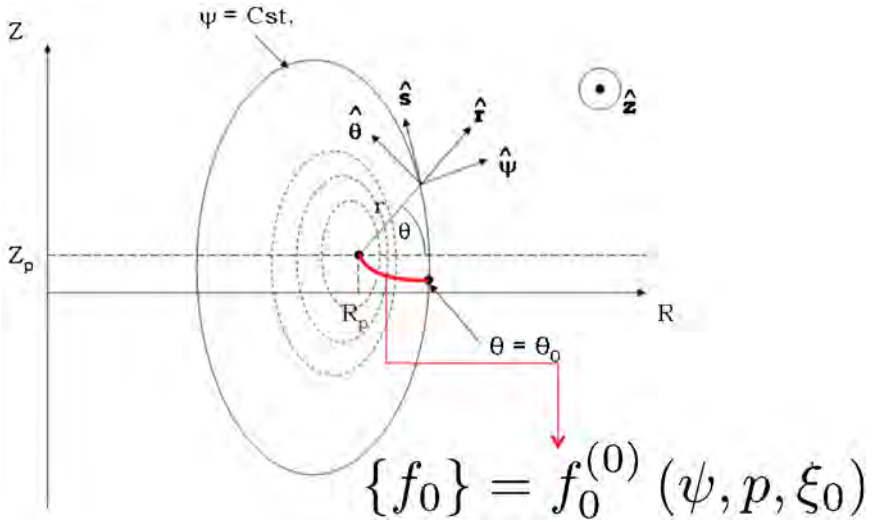


Fig. 4. Poloidal cross section of the plasma characterized by nested magnetic flux surfaces. The bounce-averaged electron distribution function $f_0^{(0)}$ is calculated on the thick line, where the magnetic field is minimum on a flux surface. The distribution function f_0 can then be determined at any poloidal angle θ using the relation $\xi(\theta, \xi_{s0})$ (Ref. 43).

resulting guiding-center Fokker-Planck equation is solved to calculate rf-driven current sources[40]:

$$\frac{\partial f_0}{\partial t} + \mathbf{v}_{gc} \cdot \nabla_{\mathbf{x}} f_0 = \mathcal{C}(f_0) + \mathcal{Q}(f_0) + \mathcal{E}(f_0) + \mathcal{T}(f_0) \,, \qquad (3)$$

where

$\mathcal{Q}(f_0) \equiv \nabla_{\mathbf{p}} \cdot (\mathbb{D}_{ql} \cdot \nabla_{\mathbf{p}} f_0)$ = quasi-linear operator describing the resonant interactions between rf waves and electrons as a diffusion process characterized by $\mathbb{D}_{ql} \propto \|\mathbf{E}_1\|^2$

$\mathcal{E}(f_0) \equiv -\nabla_{\mathbf{p}} (e \langle \mathbf{E}_2 \rangle \cdot f_0)$ = ohmic electric field operator

$\mathcal{T}(f_0) \equiv \nabla_{\mathbf{x}} \cdot (\mathbf{F}_{\mathbf{x}} f_0 - \mathbb{D}_{\mathbf{x}} \cdot \nabla_{\mathbf{x}} f_0)$ = operator describing possible anomalous spatial transport of the electrons by turbulence or MHD activity, resulting in a pinch $\mathbf{F}_{\mathbf{x}}$ and/or a diffusion $\mathbb{D}_{\mathbf{x}}$.

In the derivation of Eq. (3), the quasi-linear approximation holds for small wave field amplitude neglecting electrons trapping in the rf field itself. The guiding-center velocity can be decomposed into the dynamics along the field line and a slower drift $\mathbf{v}_D$ across the field lines as $\mathbf{v}_{gc} = p_\parallel \hat{b}/(\gamma m) + \delta \mathbf{v}_D$.[f] Here, $\mathbf{p} = p_\parallel \hat{b} + \mathbf{p}_\perp$, and the pitch-angle cosine is defined as $\xi = p_\parallel / p$. The bootstrap current can be calculated consistently with other mechanisms (rf waves, constant electric field, etc.) by solving the drift kinetic equation, taking into account $\mathbf{v}_D \neq 0$. It requires important theoretical developments discussed in Sec. VII (Refs. 41 and 42).

In axisymmetric plasmas, f_0 is a function of four coordinates, two in configuration space $\mathbf{x} \rightarrow (\psi, \theta)$ and two in momentum space $\mathbf{p} \rightarrow (p, \xi)$, where ψ is the poloidal magnetic flux and θ is the poloidal angle. The poloidal dependence may be removed by introducing the low collision ordering $\delta^2 \ll \nu^* \ll 1$, where $\nu^* = \nu_{dt} \tau_b$ is the collisionality parameter defined for trapped electrons as the ratio of the detrapping frequency ν_{dt} to the bounce frequency τ_b^{-1} (Ref. 15). The condition $\nu^* \ll 1$ implies that all electrons may fully complete their closed orbits in a poloidal cross section, before being scattered by a collision. Consequently, f_0 is conserved along the field lines, and it is sufficient to solve the kinetic equation at the poloidal position θ_0, where the magnetic field is minimal on a given magnetic flux surface ψ, $B_0(\psi) \equiv B(\psi, \theta = \theta_0) \equiv \min_\theta [B(\psi, \theta)]$,[g] as illustrated in Fig. 4. An average $\{\mathcal{O}\}$ over electron orbits is defined for any operator $\mathcal{O}$. It is an exact annihilator of the guiding-center term, i.e., $\{\mathbf{v}_{gc} \cdot \nabla_{\mathbf{x}} f_0\} = 0$. Averaging Eq. (3) over electron orbits yields the 3-D equation

$$\frac{\partial f_0^{(0)}}{\partial t} = \{\mathcal{C}(f_0)\} + \{\mathcal{Q}(f_0)\} + \{\mathcal{E}(f_0)\} + \{\mathcal{T}(f_0)\} \,, \qquad (4)$$

where $f_0^{(0)} \equiv \{f_0\}$ is a function of three coordinates (ψ, p, ξ_0), with $\xi_0 = \xi(\psi, \theta_0)$. The evolution of the

[f]Radio-frequency–driven current source calculations often assume the zero banana width approximation corresponding to $\mathbf{v}_D = 0$.

[g]The bounce-averaged electron distribution function must be calculated at the minimum of B, as it is the only point on the flux surface crossed by all orbits.

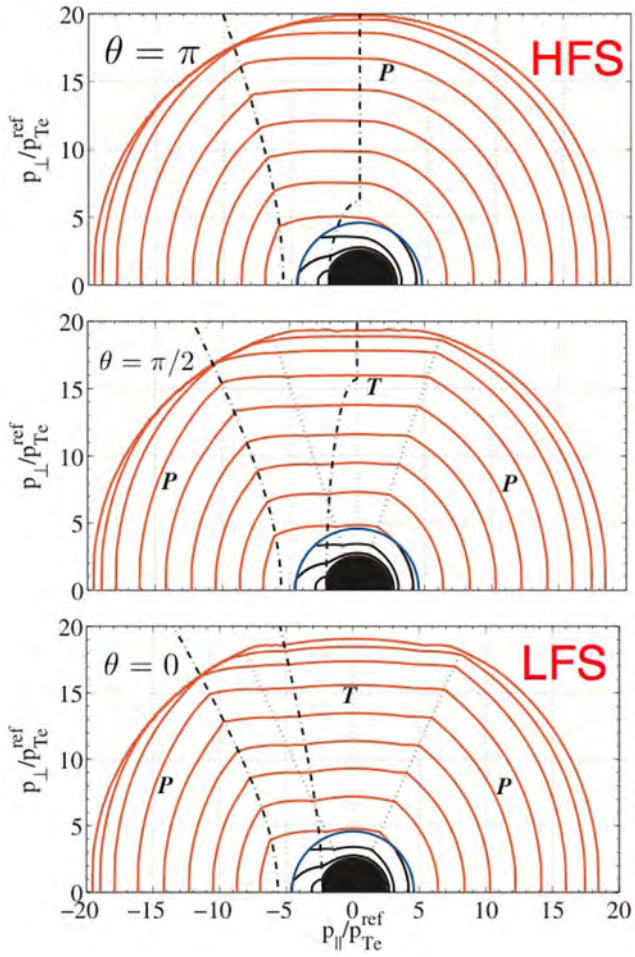


Fig. 5. Reconstruction of the non-Maxwellian electron distribution f_0 during LH current drive, at different poloidal angles θ between HFS and LFS using the relation $\xi(\theta, \xi_0)$ based on the conservation of kinetic energy and magnetic moment.[43]

guiding-center pitch angle ξ along the field lines is given by $1-\xi^2=\Psi(\psi,\theta)(1-\xi_0^2)$, with $\Psi(\psi,\theta)=B(\psi,\theta)/B_0(\psi)$. It is derived from the conservation of the kinetic energy $\mathcal{E}$, a constant of the motion, and of the magnetic moment $\mu=p_\perp^2/B$, an adiabatic invariant.[h]

Electron dynamics in the low collisionality regime is illustrated in Fig. 5, where the distribution function f at various poloidal angles θ is reconstructed from $f_0^{(0)}$ and the relationship between ξ_0 and ξ. This particular example has been used to identify the lack of symmetry between the high and low magnetic field sides (HFS and LFS in Fig. 5) of the fast electron bremsstrahlung during LH current drive experiments in the tokamak Tore Supra.[43]

[h]As calculations are restricted to the zero-orbit width limit, the magnetic flux surface label ψ is also a constant of electron motion.

The applicability of the 3-D bounce-averaged Fokker-Planck equation is restricted to the domain where $\nu^* \ll 1$ and is illustrated in Fig. 6. The divergence of ν^* as $\rho \to 0$ is a consequence of the definition of ν^* proportional to the detrapping frequency $\nu_{dt} \equiv \nu_{coll}/\sqrt{\epsilon}$. The term $\sqrt{\epsilon}$ accounts for the width of the trapped region. However, as the number of trapped particles vanishes for $\rho \to 0$, their measure becomes negligible. Thus, to the leading order, the moments of the distribution (current, power deposition) are not affected by the divergence of ν^* in the center. Near the plasma edge, however, ν^* increases as the temperature decreases, and the numerical determination of a current source in the region where $\nu^* \geq 1$ requires solving the four-dimensional Eq. (3), a much larger numerical effort.

Since collisions dominate the dynamics of thermal particles in tokamak plasmas, the distortion $\delta f_0^{(0)} = f_0^{(0)} - f_M$ from a Maxwellian represents a small fraction of the electron population. It is then possible to linearize the electron-electron collision operator as $\mathcal{C}_{e-e}(f_0^{(0)}, f_0^{(0)}) = \mathcal{C}_{e-e}(f_0^{(0)}, f_M) + \mathcal{C}_{e-e}(f_M, f_0^{(0)})$. The contribution of $\mathcal{C}_{e-e}(f_M, f_0^{(0)})$ is particularly important for accurate quantitative estimates of the rf-drive current source.[44] It is an integral term usually determined to conserve particles and momentum, but not energy, in the e-e collision operator, such that radial transport of energy is not necessary to reach a steady-state solution.

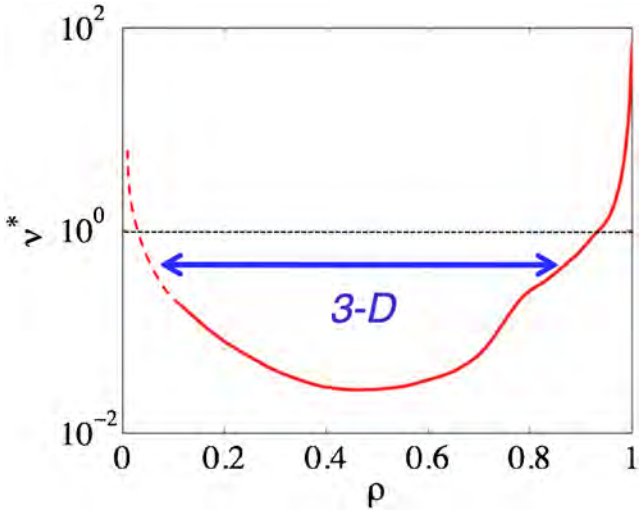


Fig. 6. Domain of validity of the low collisionality approximation for the ITER ELMy H-mode scenario. The rise of ν^* toward the plasma core arises from the $\epsilon^{-1/2}$ dependence in the detrapping frequency and does not impact the current drive calculation as the trapped region shrinks for $\rho \to 0$. Near the plasma edge, ν^* increases because of the decrease in electron temperature such that bounce-averaged Fokker-Planck calculations become invalid.[39]

III.B. Propagation and Absorption of rf Waves

The choice of numerical method to solve Maxwell's equation for rf fields is based on a spatial ordering between the wavelength λ, the beam width d, and the gradient length characterized by the minor radius a. When $\lambda \ll d \ll a$, the perturbed electric field can be expressed in the form $\mathbf{E}_1(\mathbf{x},t) \simeq \mathbf{E}_{k\omega}(\mathbf{x},t)\exp[i\mathbf{k}\cdot\mathbf{x}-\omega t]$, providing a wave front exists, i.e., $\mathbf{k}=\nabla_{\mathbf{x}}S$ and $\omega=\partial S/\partial t$, where S is the eikonal function. In the limit $\lambda \ll a$, the Wentzel-Kramers-Brillouin (WKB) approximation holds such that the background plasma can be considered as locally constant and uniform, allowing Fourier space description, the definition of a group velocity $\mathbf{v}_G$ and a local conductivity tensor $\mathbb{S}(\mathbf{x},t)$. Within the additional approximation of a well-collimated beam $\|\Delta\mathbf{k}\times\mathbf{v}_G\|/\|\mathbf{k}\times\mathbf{v}_G\| \sim \lambda/d \ll 1$, the electric field $\mathbf{E}_{k\omega}(\mathbf{x},t)$ is a smooth wavefield envelope corresponding to a narrow spectral width. Since $\|\nabla_{\mathbf{x}}\mathbf{E}_{k\omega}\cdot\mathbf{v}_G\|/\|\mathbf{E}_{k\omega}\times\mathbf{v}_G\| \ll \|\mathbf{k}\|$ and $\|\partial\mathbf{E}_{k\omega}/\partial t\cdot\mathbf{v}_G\|/\|\mathbf{E}_{k\omega}\times\mathbf{v}_G\| \ll \omega$, the wave equation may be expressed in the compact form $\mathbb{D}_{k\omega}\cdot\mathbf{E}_{k\omega} = i\nabla_{\mathbf{k}}\mathbb{D}_{k\omega} : \nabla_{\mathbf{x}}\mathbf{E}_{k\omega}$, where $\mathbb{D}_{k\omega}(\mathbf{x},t)=\mathbf{nn}-n^2\mathbb{I}+\mathbb{K}_{k\omega}(\mathbf{x},t)$ is called the dispersion tensor. The permittivity tensor is $\mathbb{K}_{k\omega}(\mathbf{x},t)=\mathbb{I}+\mathbb{X}_{k\omega}(\mathbf{x},t)$, where $\mathbb{X}_{k\omega}(\mathbf{x},t) = i\mathbb{S}_{k\omega}(\mathbf{x},t)/(\varepsilon_0\omega)$ is the susceptibility tensor. Here, $\mathbf{n}=c\mathbf{k}/\omega$ is the wave refractive index decomposed as $\mathbf{n}=n_{\|}\cdot\hat{b}+\mathbf{n}_{\perp}$, $n=\|\mathbf{n}\|$, and the polarization vector is $\mathbf{e}_{k\omega}=\mathbf{E}_{k\omega}/\|\mathbf{E}_{k\omega}\|$. The weak damping approximation $\left|\mathbb{D}^A_{k\omega}(i,j)\right|/\left|\mathbb{D}^H_{k\omega}(i,j)\right| \ll 1$ generally holds for propagating rf waves in tokamak plasmas, where $\mathbb{D}^H_{k\omega}$ and $\mathbb{D}^A_{k\omega}$ are the Hermitian and anti-Hermitian parts of $\mathbb{D}$, respectively. Considering the small parameter $\delta_D = \max\left[\left|\mathbb{D}^A_{k\omega}(i,j)\right|/\left|\mathbb{D}^H_{k\omega}(i,j)\right|\right]$, and assuming that both $n_{\|}$ and ω are real numbers since the rf wave propagates, it is possible to perform an expansion in powers of δ_D (Ref. 45). To order zero, the wave equation reduces to $\mathbb{D}^H_{k\omega}(n_{\perp 0}, n_{\|}, \omega)\cdot\mathbf{e}_{k\omega 0}=0$, and propagating eigenmodes satisfy the dispersion relation $\det\mathbb{D}^H_{k\omega}(n_{\perp 0}, n_{\|}, \omega) = \mathcal{D}(n_{\perp 0}, n_{\|}, \omega)=0$. To the next order in δ_D, the equation of energy transfer between the rf wave and the electrons is obtained, which involves $\mathbb{D}^A_{k\omega}$. The ordering resulting from the weak damping approximation leads to the separation between the propagation and the absorption of the rf waves, which represents a considerable simplification for numerical calculations. The determination of the rf power flow trajectory is determined by solving the ray equation $\mathcal{D}=0$ (Ref. 16). Since $\mathbb{D}^H_{k\omega}$ is related to the principal value of $\mathbb{S}_{k\omega}$, $\mathbb{D}^H_{k\omega}(f_0^{(0)}) \simeq \mathbb{D}^H_{k\omega}(f_M)$ and the distortion of the distribution function due to the resonant absorption of the rf waves with electrons can be neglected in the calculation of ray trajectories. This approximation simplifies the calculations considerably, since ray paths can be determined independently of the distribution function. Conversely, $\mathbb{D}^A_{k\omega}$, which appears only in the equation of absorption, is related to the resonant part of $\mathbb{S}_{k\omega}$ according to the condition of resonance $\gamma - n_{\|}p_{\|}/(mc) - n_{\Omega}\Omega/\omega$, where n_{Ω} is a cyclotron harmonic mode number. As a consequence, $\mathbb{D}^A_{k\omega}(f_0^{(0)}) \neq \mathbb{D}^A_{k\omega}(f_M)$, and the consistency between rf-wave damping calculations and $f_0^{(0)}$ is essential. A convergence procedure, namely, the quasi-linear self-consistency, is followed to solve iteratively the Fokker-Planck equation consistently with the power flow along each ray.[46]

When $\lambda \sim d \ll a$, diffraction effects are important, and a more complex set of equations called beam tracing must be solved, while the weak damping approximation still holds such that the propagation and damping of rf waves can still be determined separately.[47] If $d \sim a$, the structure of the wavefield $\mathbf{E}_1(\mathbf{x},t)$ may not be cast in the simple form used for the ray tracing, and full-wave calculations of the Maxwell equation must be performed, regardless of the ordering between λ and d. In this case, the concept of propagation must be replaced by a description in terms of modes.[48–50] As a consequence, the Fokker-Planck equation describing the exchange of energy between the electrons and the rf waves must be solved simultaneously with the field equations, which represents a considerable computational effort for determining the rf-driven current source. Such an approach has been made possible recently for the LH wave[51–53] owing to improved computing resources.

IV. C3PO/LUKE, A rf CURRENT SOURCE MODULE FOR INTEGRATED MODELING

From the set of approximations described in Sec. III, the generic structure of a rf current source module for integrated tokamak modeling illustrated in Fig. 7 has been developed.

IV.A. C3PO Ray-Tracing Code

Ray equations are solved by the versatile 3-D ray-tracing code C3PO with a design based on chain derivatives for maximum modularity regarding the choice of coordinate system and dielectric model.[16] A coordinates system $(\rho(\psi),\theta,\phi)$ reflecting the nested structure of magnetic flux surfaces in tokamaks[i] is used for fast and accurate calculations inside the plasma separatrix. Analytical derivatives are used on a vectorized form of the discrete numerical two-dimensional axisymmetric toroidal MHD based on Fourier series expansion and piecewise cubic interpolation using Hermite polynomials. A 3-D perturbation due to the discreteness of the magnetic coil system or plasma fluctuations can be incorporated in the calculations, provided the magnetic flux surfaces remain nested.

[i]The code may be used for any axisymmetric magnetic configuration (cylindrical, toroidal, dipolar).

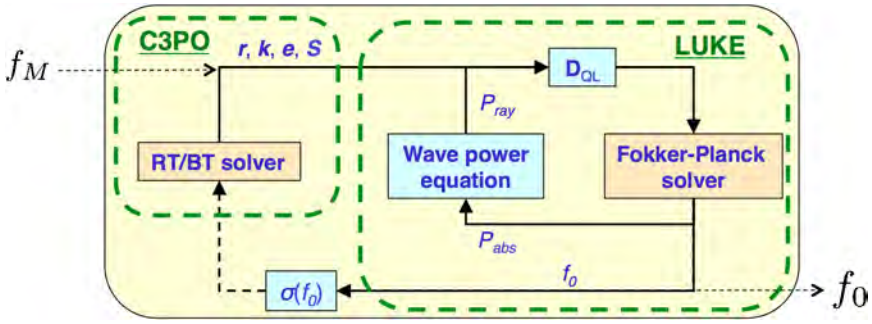


Fig. 7. Schematic view of the current drive module based on the C3PO ray-tracing code and the 3-D linearized relativistic bounce-averaged electron Fokker-Planck solver LUKE (Refs. 15 and 16). If the weak damping approximation fails, the plasma conductivity must be calculated consistently between the propagation and the absorption of the rf wave. Such a scheme is considered when the wave propagation is calculated by a full-wave code.

The set of six ordinary differential equations is solved by a (4, 5) Runge-Kutta-Fehlberg method, and the calculation of the ray path is stopped a few steps after the rf power is linearly absorbed, to account for the possible reduction of the damping rate by the quasi-linear effects arising from the flattening of f_0 in the direction of diffusion where the kinetic resonance takes place. Initial conditions are calculated at the plasma separatrix, and specular reflections of the rays may be enforced at $\rho = 1$ if needed, depending upon the characteristics of the rf wave. The physics of the rf waves outside the separatrix is described by external dedicated modules, like the reflections of EC waves on the metallic wall, or antenna coupling for the LH wave.[54]

Written in C language, C3PO is incorporated as a mex-file in the flexible Matlab computational environment used by the rf current source module.[55] It takes advantage of the distributed and remote computing capability designed for this module, such that calculations involving multiple rays may be performed simultaneously on separate processors.

IV.B. LUKE Fokker-Planck

The numerical resolution of the 3-D linearized relativistic bounce-averaged electron guiding-center Fokker-Planck equation requires it to be cast in a flux conservative form of the kinetic equation, such that a two-grid technique can be applied, one for the fluxes and another for the distribution itself.[15] Conditions at the internal and external boundaries of the integration domain are naturally set on the flux grid to ensure the conservation of particles in $f_0^{(0)}$. The general form of the equation is $\partial f_0^{(0)}/\partial t + \nabla\cdot\mathbf{S}^{(0)} = s_+^{(0)} - s_-^{(0)}$, where $s_\pm^{(0)}$ are possible sources and sinks,[j] and the differential operator[40]

$$\nabla\cdot\mathbf{S}^{(0)} = \frac{B_0}{\tilde{q}\lambda_o}\frac{\partial}{\partial\psi}\left(\frac{\tilde{q}\lambda_o}{B_0}\|\nabla\psi\|S_\psi^{(0)}\right) + \frac{1}{p^2}\frac{\partial}{\partial p}\left(p^2 S_p^{(0)}\right) - \frac{1}{\lambda_o p}\frac{\partial}{\partial\xi_0}\left(\lambda_o\sqrt{1-\xi_0^2}S_\xi^{(0)}\right) \quad (5)$$

incorporates the dynamics in both momentum and configuration spaces, where $\lambda_o(\psi,\xi_0)$ is the bounce time normalized to the transit time of the electrons with parallel momentum only[k] and $\tilde{q}(\psi)$ is a magnetic field form factor. In Eq. (5), $\mathbf{S}^{(0)} = -\mathbb{D}^{(0)}\cdot\nabla f_0^{(0)} + \mathbf{F}^{(0)} f_0^{(0)}$, where the diffusion pseudotensor $\mathbb{D}^{(0)}$ and convection pseudovector $\mathbf{F}^{(0)}$ are the sum of all physical contributions of interest: rf-wave quasi-linear interaction, coulomb collision, constant electric field acceleration, radial transport of fast electrons, etc.

The Fokker-Planck code LUKE uses the same coordinate system as C3PO, which minimizes numerical inaccuracies in the coupling between the two codes. It solves the Fokker-Planck equation by a finite difference technique. All grids may be nonuniform, allowing a refined mesh at a requested location of phase-space—such as the interaction domain between electrons and rf waves—without an excessive computational cost. Cross derivatives are expressed such as to satisfy boundary conditions intrinsically, ensuring a stable numerical scheme, even when the rf diffusion is large. The interpolation between the flux and distribution grids is performed using the Chang and Cooper method for the momentum grid[56] and linear interpolations for the radial and pitch-angle grids. A fully implicit time scheme is used by default, as it is stable for a large integration time step Δt, but a Crank-Nicholson time scheme is also implemented if a more accurate time evolution of the distribution function is needed. The matrix to be solved has a block structure corresponding to the momentum dynamics on each magnetic flux surface,

[j] $s_\pm^{(0)}$ can be used to describe physical effects that are beyond the approximations of the Fokker-Planck equation, such as a loss of fast electrons trapped in the magnetic ripple, or the generation of runaway electron through avalanche mechanisms.[91–93]

[k] $\lambda_o(\psi, \pm 1) = 1$

with complex bands of nonzero elements linking neighboring blocks and describing the spatial dynamics of magnetized electrons, including trapping and detrapping consistent with radial transport, as shown in Fig. 8. It is nevertheless highly sparse, and generalized incomplete lower-upper factorizations are valuable techniques to solve iteratively the corresponding set of linear equations. Written also in Matlab, LUKE can use built-in linear system solvers but usually relies on more powerful external matrix inversion packages such as MUMPS or PETSc (Refs. 57 and 58). Parallelization and remote computing are incorporated in LUKE to reduce calculation time.

IV.C. Link Between C3PO and LUKE

The link between C3PO and LUKE is crucial in rf-driven current source calculations.[46] The bounce-averaged quasi-linear diffusion tensor $\mathbb{D}_{\mathrm{ql}}^{(0)}(\psi,p,\xi_0)=\sum_j \mathbb{D}_{\mathrm{ql},j}^{(0)}(P_j^{\mathrm{rf}},\psi,p,\xi_0)$ is determined on each magnetic flux surface ψ by summing over the contribution of all rays in the associated incremental volume $\Delta V(\psi)$, including multiple passes of the same ray across a given flux surface. The determination of individual contributions $\mathbb{D}_{\mathrm{ql},j}^{(0)}$ from each ray element to the quasi-linear diffusion coefficient is described in detail in Ref. 59. The fully relativistic quasi-linear operator derived by Lerche[60] is applied to the specific case of a localized beam with a Gaussian transverse power profile. It is transformed to be applicable to nonuniform toroidal plasmas. In particular, the poloidal location of the wave-particle interaction is accounted for in the resonance condition to account for the variations of electron momentum along its orbit due to the magnetic mirror effect.

The power flow P_j^{rf} in the ray fragment depends on the integrated quasi-linear damping rate of the ray from launching to the ray fragment j as $dP_j^{rf}/dV=P_j^{abs}$, as shown in Fig. 7. Since the power P_j^{abs} absorbed by the electrons on each magnetic flux surface for each fragment j is a moment of the distribution function $f_0^{(0)}$solution to the Fokker-Planck calculations, the calculations of $\mathbb{D}_{\mathrm{ql}}^{(0)}$ and $f_{,0}^{(0)}$ are interdependent. Starting from the linear damping rate, the quasi-linear convergence between $\mathbb{D}_{\mathrm{ql}}^{(0)}$ and $f_0^{(0)}$ is carried out following the power flow along each ray until both quantities become consistent with a relative accuracy $<1\%$. Convergence of this highly nonlinear interdependent system can be reached in a reasonable time if a reduced number of rays is used. For the LH wave, one ray is used per poloidal antenna row and per significant positive or negative lobes in the launched power spectrum, in accordance with the prescription of Cairns and Fuchs.[61] For the EC wave, a set of beamlets derived from Gaussian optics in vacuum are considered, neglecting the plasma effects on the paraxial approximation. This approach is consistent with the Fourier theory on which the wave description is based, as the initial spectral width associated with each ray relates to the beam width.

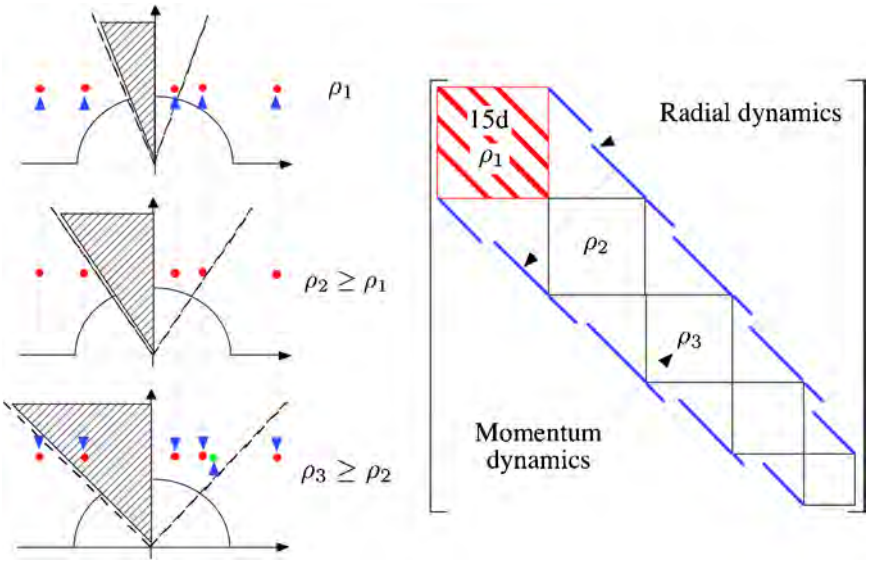


Fig. 8. Electron dynamics in momentum and configuration spaces, and the corresponding form of the matrix to be inverted. Each block describing the dynamics in momentum space is made of 15 diagonals. The diagonal linking neighboring blocks corresponds to the radial transport, which may trap or detrap electrons, depending upon their radial position and perpendicular velocity. The fast electron transport across magnetic flux surfaces in the present version of the LUKE code is performed at fixed pitch angle, since only $D_{\psi\psi}$ and F_ψ are considered to describe an ad hoc radial diffusion and convection.[15] Consequently, the magnetic moment is not conserved in this process.

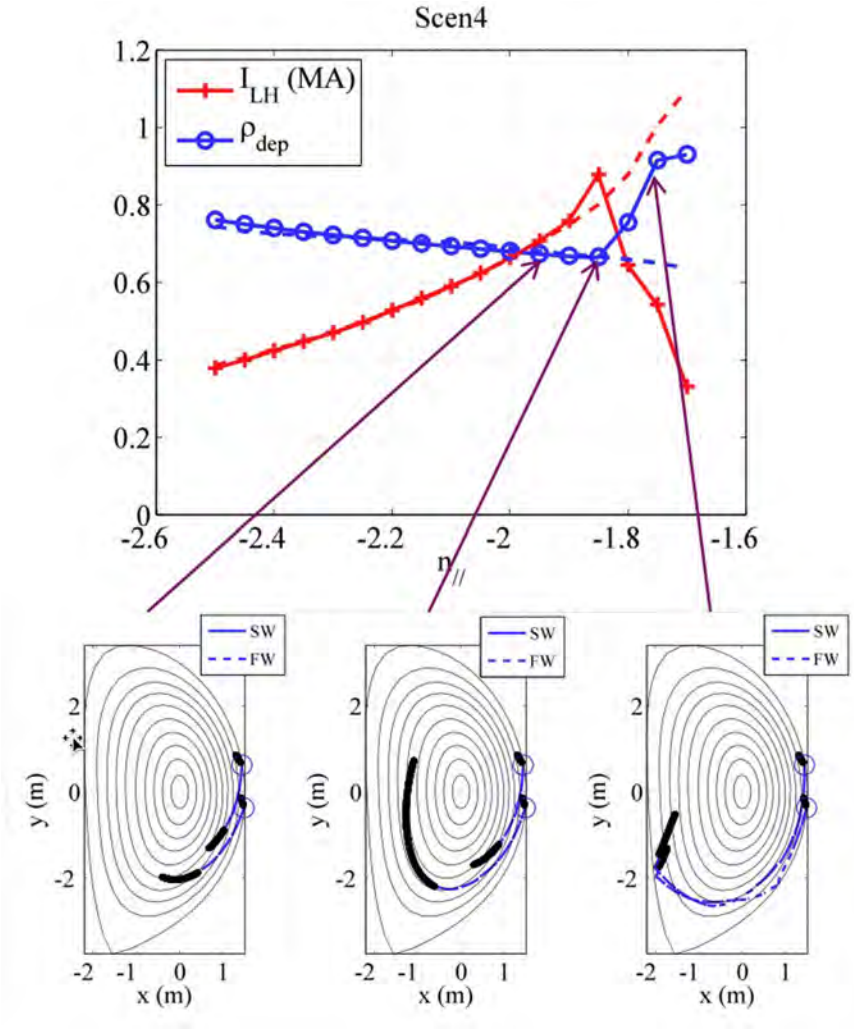


Fig. 9. Total driven current and peak location of power deposition calculated by C3PO and LUKE. The LH wave propagation is represented by four rays for the three different cases $n_{\|0} = -1.95$, $n_{\|0} = -1.85$, and $n_{\|0} = -1.75$. The thicker ray portions identify regions of strong power absorption. The evolution of the ray polarization is also indicated (SW for slow wave and FW for fast wave).[22]

V. NUMERICAL SIMULATIONS OF THE CURRENT DRIVEN BY LH WAVES

The ray-tracing code C3PO and the Fokker-Planck code LUKE have both been extensively benchmarked against analytical and numerical results before being used together for realistic current drive modeling.[26,43,62–66] For example, in the low power limit when $\mathbb{D}_{ql}^{(0)} \sim 0$, the wave damping calculated by LUKE on a Maxwellian distribution agrees with relativistic linear theory, which is an essential validation of the code.[46]

The current source module based on C3PO and LUKE has been used to evaluate in detail the capability of the LH wave for driving a current in the tokamak ITER in an ELMy H-mode regime characterized by a steep pressure gradient at the plasma edge.[22] With 20 MW of LH power launched at 5 GHz by a passive active multijunction characterized by a directivity of 70%, a maximum driven current of 0.9 MA is predicted for a launched parallel refractive index $n_{\|0} \simeq -1.85$, as shown in Fig. 9. The power deposition profile is very narrow and peaks around $\rho \simeq 0.7$. Given the far off-axis location of power deposition, trapped electrons are found to play an important role and reduce the current drive efficiency by 25%. For $|n_{\|0}| > 1.85$, the current drive efficiency is a monotonic decreasing function of $|n_{\|0}|$, in full agreement with Fisch's current drive theory. The regular parametric dependence with $|n_{\|0}|$ is the direct consequence of linear single-pass absorption of all rays in the absence of a spectral gap, given the high electron temperature.[1] In this

[1]The spectral gap is defined as the difference $n_{\|LD} - n_{\|0}$ between the initial $n_{\|0}$ prescribed by antenna geometry and waveguide phasing, and the damping condition $n_{\|LD}$ (Ref. 83).

case, all the assumptions used for deriving the equations solved by the codes used in the current source module are valid. When $|n_{\|0}|$ is lowered below its optimal value, the Stix-Golant accessibility condition restricts the penetration of the wave in the plasma, and the rays experience multiple reflections at the plasma edge before being absorbed. In this case, the current drive efficiency decreases rapidly with $|n_{\|0}|$, and the absorption remains localized near the separatrix, in a cold region of the plasma with large gradients where both the low collisionality and eikonal approximations may fail. Increasing the launched LH power above 20 MW improves the LH current drive efficiency significantly, and the wave can penetrate deeper in the plasma before being absorbed. Such dependence indicates unambiguously that the quasi-linear plateau is only partially formed and the saturation regime is not yet reached (Fig. 10).

The simulation of LH current drive in existing tokamaks is often much more challenging as colder electron temperatures result in a larger spectral gap. Based on antenna geometry and the respective phasing of waveguides, an initial spectrum at the antenna mouth can be estimated analytically or calculated with additional details by a variety of numerical codes such as GRILL (Ref. 67), ALOHA (Ref. 68), and TOPLHA (Ref. 69). This power spectrum is made of a number of individual lobes with an approximately Gaussian shape. It is shown that only the main lobes in the forward and backward directions, which peak at $n_{\|0}^{+}$ and $n_{\|0}^{-}$ with respect to the magnetic field, contribute significantly to LH current drive.[22,70,71] Other satellite lobes have a negligible contribution. The number of rays in LH current drive calculations with C3PO/LUKE is given by the triple product between the number of main lobes in the forward and backward directions, the number of horizontal rows in the waveguide array with the corresponding initial poloidal positions for the rays, and the number of antennas. In such a way, all rays are spectrally decoupled, a condition of applicability of the WKB approximation.[8,61] With this prescription, the number of rays remains low enough for a reasonably fast quasi-linear numerical convergence, as discussed in Sec. IV (Ref. 46).

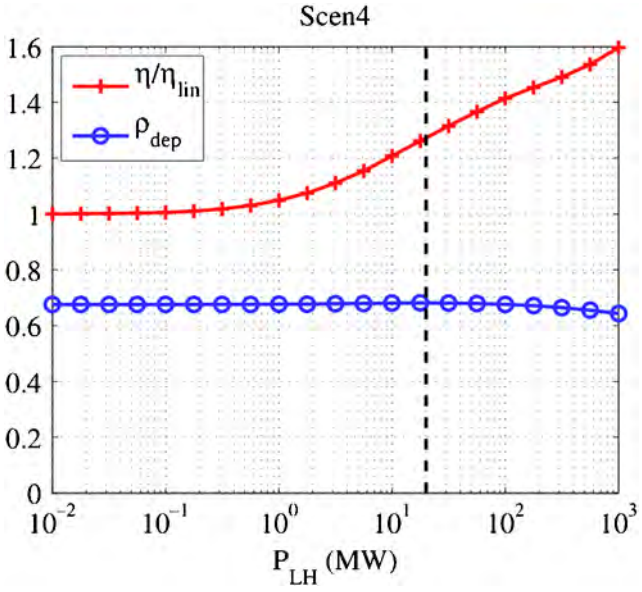


Fig. 10. Variations of the location of power deposition ρ_{dep} and the normalized efficiency η/η_{lin} as a function of the total power P_{LH}. The reference η_{lin} is the value obtained for $P_{LH} = 0.01$ MW. The vertical dashed line indicates the value $P_{LH} = 20$ MW (Ref. 22).

Quantitative ray-tracing and Fokker-Planck simulations of LH current drive experiments in C-MOD and Tore Supra tokamaks have recently shown that these tools are able to reproduce reasonably the experimental phenomenology when the spectral gap is small enough, i.e., when ray absorption occurs in a few passes (low-density and high-temperature plasmas). In this case, the reconstruction of the hard X-ray emission from fast electron bremsstrahlung tomography,[43,72,73] which provides a detailed insight of the electron distribution function, is in good agreement for observed results.[71,74,75] When the spectral gap becomes important, stochasticity develops in the ray propagation long before the power carried by the wave can be damped on the thermal electron population.[76] An interesting example is given for the 6-min discharge sustained in the Tore Supra tokamak with 3 MW of LH power launched by two LH antennas. As shown in Fig. 11, small modifications of the ray initial conditions completely change the predicted absorption, a typical situation when the code is used well beyond its domain of validity.[65] In this case, even for a discharge in steady state, the predicted current fluctuates considerably in relation to minor variations of the reconstructed toroidal MHD equilibrium at different time slices, in full contradiction with experimental observations. This problem, which was identified a long time ago, remains unsolved so far despite considerable theoretical and numerical efforts following many experimental investigations.[77–79]

The possible role of electron density fluctuations at the plasma edge on LH wave propagation and absorption was pointed out a long time ago[76,80,81] and revisited recently[65,82] to assess if the time-averaged solution in large spectral gap regimes could be robust with respect to small changes of plasma or wave parameters, and in good agreement with experimental observations. This approach was a physical alternative to the multiray numerical technique introduced for the same purpose.[77] As shown in Fig. 12, fluctuations resulting from the electron drift wave model lead only to a small inward radial shift of the power absorption profile, without a significant broadening. The location where the profile peaks remains therefore far from the experimental location obtained from fast electron

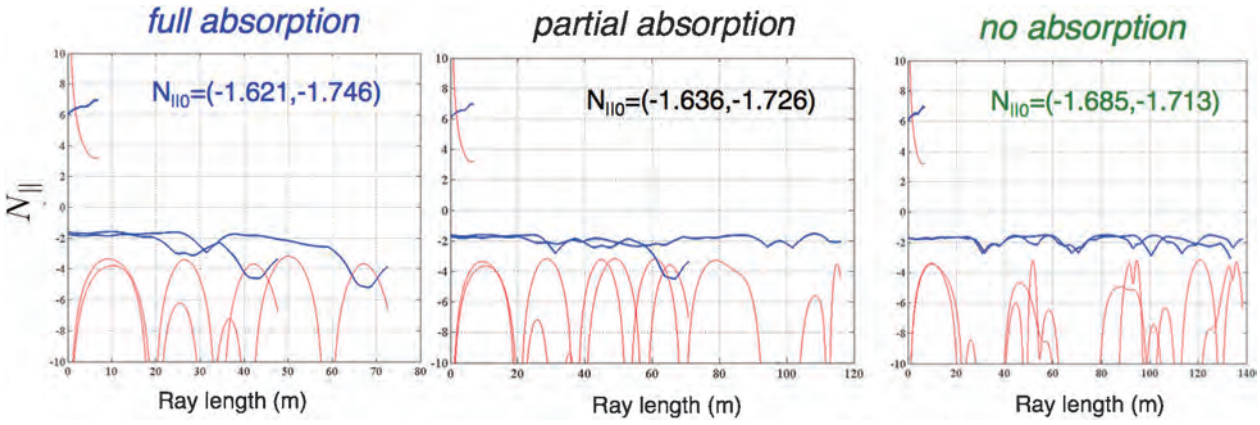


Fig. 11. Evolution of the parallel wave refractive index $n_\parallel$ along the ray trajectories (blue lines) using slightly different initial conditions, highlighting the development of the chaotic behavior of the ray dynamics long before the ray absorption (Tore Supra discharge #32299). Such a situation illustrates the large spectral gap problem. Full absorption of the LH wave takes place when the upshift exceeds locally the condition $n_\parallel \geq n_{\parallel L}$ (red lines). (a) $n_{\parallel 0} = (-1.621, -1.746)$ leads to full absorption of both rays, (b) $n_{\parallel 0} = (-1.636, -1.726)$ leads to the absorption of one ray only, and (c) $n_{\parallel 0} = (-1.685, -1.713)$ yields no absorption. Note that rays launched at $n_{\parallel 0} = 6$ are always fully absorbed by the plasma.[65]

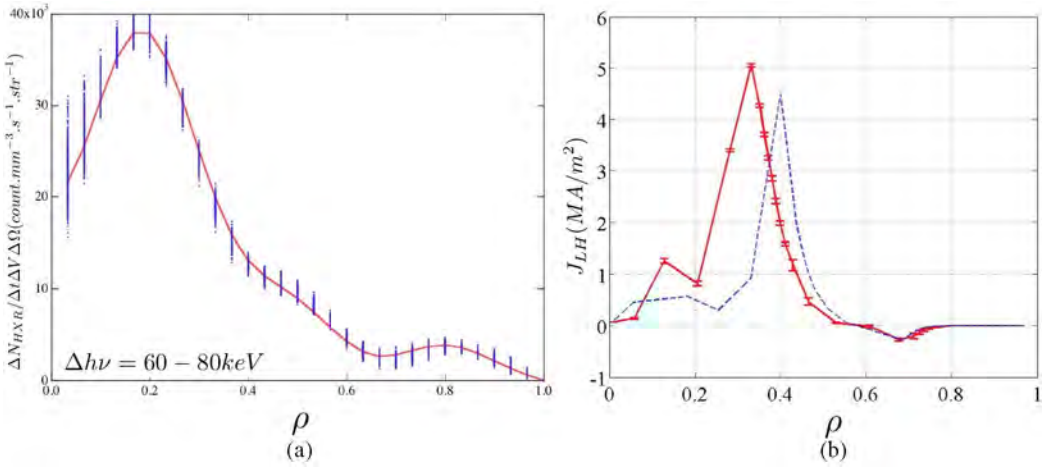


Fig. 12. Tore Supra full LH current drive discharge #32299. (a) Experimental Abel inverted profile of the fast electron bremsstrahlung in the photon energy range 60 to 80 keV, averaged between $t = 100$ s and $t = 110$ s. The sampling time is 16 ms. (b) Simulated current density with (full curve) or without (dashed curve) edge electron density fluctuation. The positive peaks correspond to the full absorption of the rays with initial conditions $N_{\parallel 0} = \{-1.73, -1.69\}$, while the small negative one to the ray $N_{\parallel 0} = 6$ launched in the opposite direction with respect to the toroidal magnetic field. Note that the two positive peaks may not be ascribed specifically to one of the two $N_{\parallel 0}$ values, because of the ray stochasticity and their interplay in the quasi-linear absorption process.[65]

bremsstrahlung tomography. In addition, the predicted LH driven current density profile remains sensitive to initial ray conditions, which indicates that these kinds of edge density fluctuations do not average out the chaotic behavior of the rays.[65]

A striking feature of the LH current drive physics is that the spectral gap is always filled, even if the toroidal refraction, usually considered as the dominant mechanism, fails, in particular, in tokamaks whose aspect ratio is very large.[83] From full-wave/Fokker-Planck simulations of TRIAM-1M LH full current drive experiments, it was suggested that the spectral gap must be already filled at the plasma edge,[50] a hypothesis that also seems valid for high-density LH current drive experiments in Tore Supra from quantitative simulations performed by C3PO/LUKE codes.[84]

VI. EFFECT OF ELECTRON DENSITY FLUCTUATIONS ON THE EC CURRENT DRIVE

Local control of the current profile using rf waves has many important applications in a fusion reactor. Among them, the real-time stabilization of $3/2$ NTMs is particularly important to maintain optimal performances of the discharge.[85] For this purpose, a source of current must be accurately driven in an island estimated to be ~ 7 cm wide in ITER. Such control has been demonstrated in present-day experiments by depositing EC wave power in a very narrow region of the plasma.[86,87] Yet, the large size of the plasma in ITER increases the distance between launching mirrors and the target to $L \simeq 120$ cm, such that even small amplitude fluctuations of the electron density near the separatrix could lead to important transient misalignment between the rf beam and the narrow width island, thereby reducing significantly the stabilizing capability of the EC waves. This detrimental effect resulting from the geometrical configuration is amplified by the use of O-1 electron cyclotron resonance heating ($\omega/2\pi = 170$ GHz) in ITER, which will operate close to the critical density where the wave propagation is particularly sensitive to density variations.[65]

A detailed investigation of the role played by density fluctuations on the EC-driven current has been recently carried out with the codes C3PO and LUKE (Refs. 65 and 88). Since the lifetime $\tau_w \sim L/c$ of an EC wavepacket from the mirror to the damping region is much shorter than all other characteristic timescales $(\tau_o, \nu_{coll}^{-1}, \tau_E)$, including the fluctuation correlation time τ_f, it is natural to consider an instantaneous response of the EC beam propagation and damping to changes in the fluctuating toroidal MHD equilibrium or distribution function. A consequence of the ordering $\tau_w \ll \nu_{coll}^{-1}$ is that the self-consistency between the wave damping rate and the distribution function $f_0^{(0)}$ must be satisfied at any given time. On the other hand, since the characteristic evolution time ν_{coll}^{-1} of the distribution function may be of the order of or longer than τ_f, the distribution function has a memory of previous states in the fluctuating equilibrium. Thus, when $\tau_f \le \nu_{coll}^{-1}$, a correct statistical description of the effect of fluctuations on $f_0^{(0)}$ cannot be obtained by calculating the time-asymptotic solution to the Fokker-Planck equation for a series of independent fluctuating equilibrium states. Instead, the fluctuating equilibrium and $f_0^{(0)}$ must evolve in a consistent way. An important consequence of the $\tau_f \le \nu_{coll}^{-1}$ ordering is that the problem of EC current drive in a fluctuating plasma requires the use of a proper solver for the time-dependent Fokker-Planck equation. The evolution time step Δt for the Fokker-Planck equation is chosen such that $\tau_f \le \Delta t$. With this condition, the fluctuating equilibrium states at two consecutive times are decorrelated and can thus be generated randomly. Since the total current density profile is almost frozen at timescales much less than τ_r^* by the large energy stored in the poloidal magnetic field, any change in the EC-driven source of current is instantaneously compensated by a local electric field. While the EC power absorption profile fluctuates on the timescale τ_f, the current density profile is constant and results from a statistical average.

Using a very simple model for the electron density fluctuations at the plasma edge,[m] it has been shown that this mechanism improves the consistency between experimental observations and quantitative predictions in the TCV tokamak where the toroidal current $I_p \simeq 150$ kA is driven entirely by EC waves.[65,88] Just like an anomalous fast electron radial transport, the fluctuations can be included in the model to broaden the predicted current density profile and reduce the driven current to the level of experimental measurement. Unlike the radial transport mechanism, however, fluctuations maintain the predicted fast electron bremsstrahlung close to the observations.[88] With density fluctuations, fast electrons are generated at radial positions that vary rapidly with time when compared to the characteristic collision time. With radial transport, fast electrons are generated at fixed radial positions and then diffuse radially while slowing down at the same time.

With the same model, the $3/2$ NTM stabilization in ITER using 20 MW of EC power has been investigated for the ELMy H-mode scenario.[65,88] As shown in Fig. 13, the amount of the EC-driven current remains almost unchanged, which is not surprising as quasi-linear effects are weak for off-axis EC current drive in ITER. However, the width of the current profile increases significantly with density fluctuation level, making $3/2$ NTM stabilization challenging if relative fluctuations at the edge exceed 10% (Fig. 14). Thus, edge density fluctuations could be a serious issue for the control of NTMs in ITER.

VII. ADVANCED DEVELOPMENTS FOR CURRENT DRIVE SIMULATIONS

For accurate calculations of current sources in the presence of steep pressure gradients, it is necessary to consider the exact orbits of charged particles in the inhomogeneous magnetized plasma as shown in Fig. 15 (Refs. 40, 89, and 90). A parameterization of the guiding-center orbits of the form $\left(\bar{\psi}, p, \xi_0\right)$ is still possible, where $\bar{\psi} \equiv cP_\phi/e = \psi - \rho_\parallel B_\phi$ is a magnetic flux-surface label and $\xi_0 \equiv \sigma\sqrt{1-\mu B_0\left(\bar{\psi}\right)/\mathcal{E}}$ is a pitch-angle coordinate. Here, P_ϕ is the generalized toroidal canonical momentum, $\rho_\parallel = p_\parallel/(ZeB)$ is the parallel Larmor radius, Z is the

[m]Fluctuations, highly localized near the separatrix, are supposed to be driven by electron drift waves for which $\tilde{k}_\parallel \sim 0$. The mode numbers used for describing the fluctuation power spectrum are restricted to values such that the eikonal approximation holds in the rf wave description.

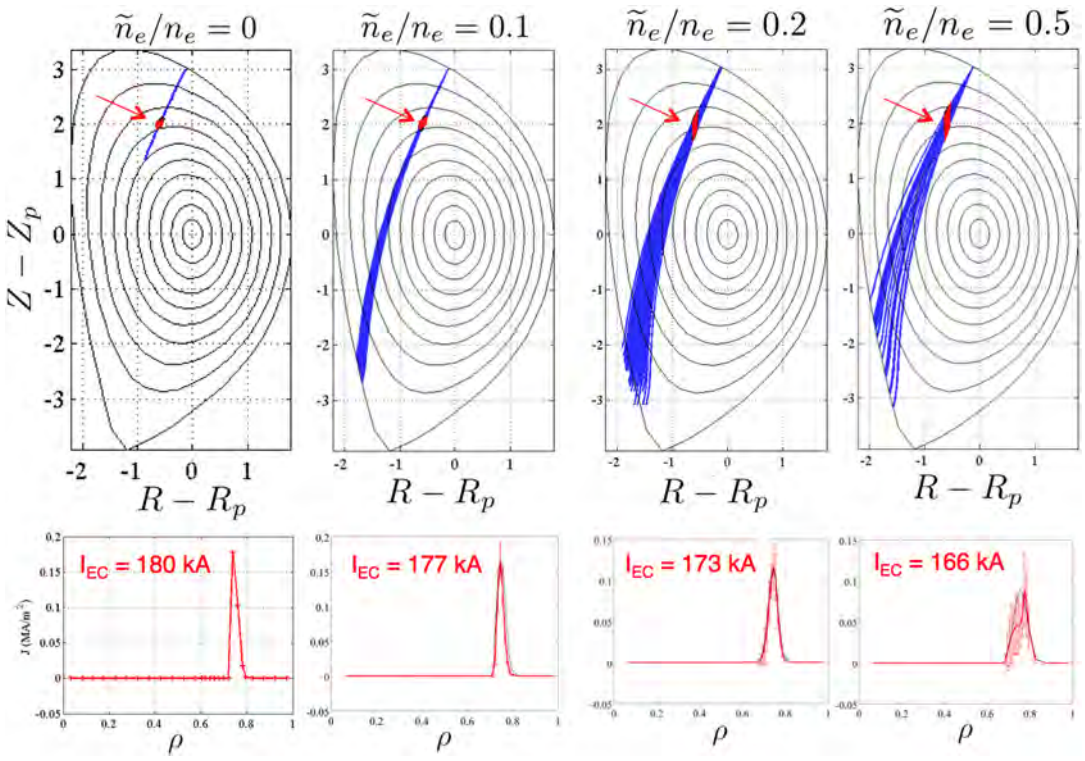


Fig. 13. ITER ELMy H-mode scenario. Poloidal cross section of the simulated discharge with EC ray trajectories and corresponding time-averaged current density profile in the stationary regime. Here, R_p and Z_p are, respectively, the major radius and the vertical position of the magnetic axis. The arrows indicate where the EC power absorption takes place, depending upon the relative amplitude of the density fluctuations. For all cases shown here, the fluctuating width is 2 cm (Ref. 65).

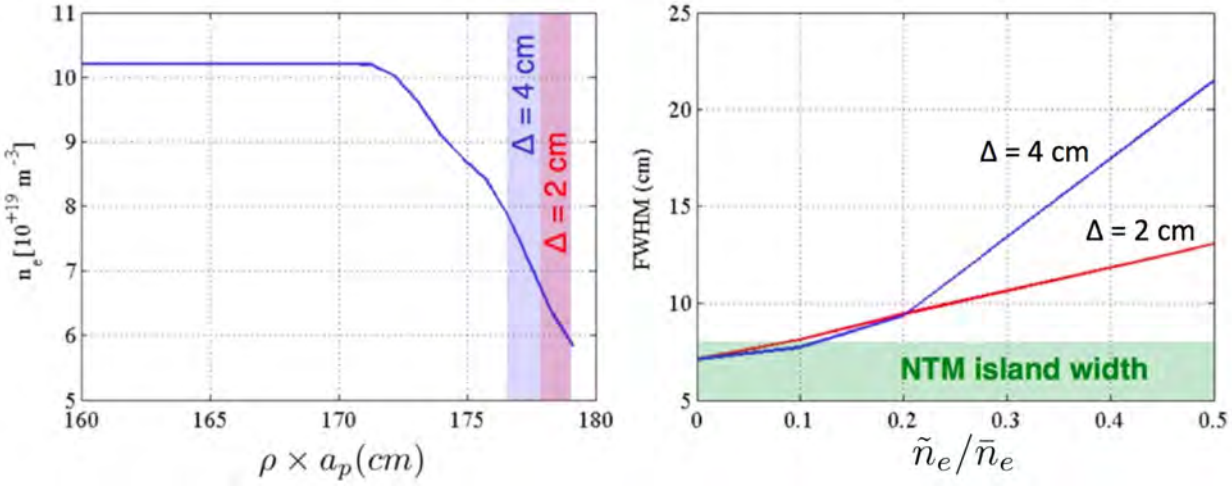


Fig. 14. ITER ELMy H-mode scenario. (a) Electron density profile in the pedestal. The shaded areas correspond to the region where fluctuations are supposed to take place. (b) The full-width at half-maximum of the EC current density profile for two widths of the fluctuating region, as a function of the edge fluctuation standard deviation. The shaded area corresponds to the estimated size of the 3/2 NTM island.[65]

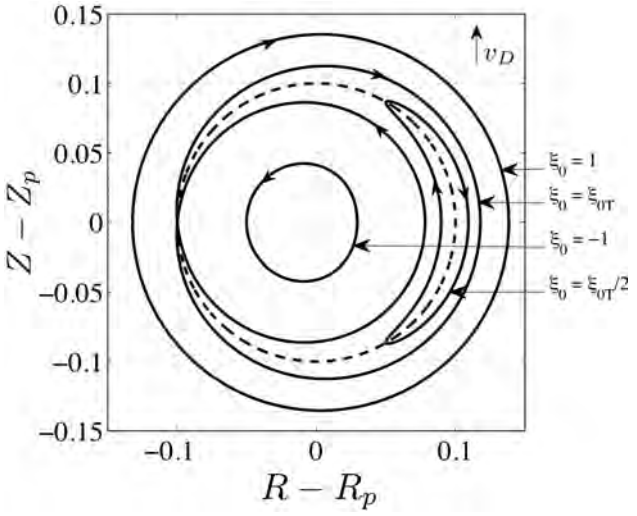


Fig. 15. Schematic representation of guiding-center orbits. The flux surface of reference $\bar{\psi}$ is drawn as a dashed line.[40]

particle charge, B_ϕ is the toroidal component of the magnetic field in the covariant representation, $\sigma = 1$ for trapped particles, and $\sigma = \mathrm{sgn}(v_\parallel)$ for circulating particles. One of the significant advantages of this representation over others in the perspective of a numerical implementation is that the trapped/passing boundary remains independent of p. In addition, it allows most of the numerical framework developed in the code LUKE to be kept in the zero-orbit width limit. Using this new generalized coordinate system, an orbit-averaged guiding-center Fokker-Planck operator is obtained in the low collisionality regime, which may be cast into the appropriate conservative form for numerical calculations when the thin orbit approximation holds. For an isotropic collision operator, explicit expressions of the convection vector and diffusion tensor are derived, and analytical expressions of the bootstrap current are obtained in the limit $Z \gg 1$ (Lorentz limit).[40] This result validates the theoretical approach, paving the way for a consistent numerical calculation of the bootstrap current in the presence of rf waves. Other cross effects between momentum and configuration spaces may also be evaluated, such as the Ware pinch in the presence of a constant electric field, and the particle transport induced by the rf waves. Including orbit effects also extends the code application of the description of ion dynamics.

VIII. CONCLUSIONS AND PROSPECTS

The toroidal current density profile plays a central role in tokamak operation, and the calculation of current sources by rf waves is essential in integrated tokamak modeling. The strong interplay between transport properties of the plasma and the current density profile in advanced steady-state regimes puts even higher constraints on simulations for accurate predictive calculations of such scenarios. The current generation and the plasma response are characterized by multiple time and space scales, whose separability have important consequences on the structures of the numerical codes. To reduce computational efforts, averages are introduced to reduce the dimensionality of the problem, which in turn restricts the range of applicability of current drive simulations. A strict numerical implementation of standard models designed for a consistent description of the dynamics between rf fields and electrons has led to the emergence of a fast, robust, and powerful tool for calculating the current driven by rf waves in magnetized plasmas. By an appropriate choice of coordinate system and a self-consistent derivation, the ray-tracing codes C3PO and the Fokker-Planck solver LUKE are particularly suitable for modular current drive calculations in integrated tokamak modeling. The power of the Matlab language and its compactness have also contributed to this result. This versatile tool can be used beyond tokamak physics for all axisymmetric magnetic configurations with nested magnetic flux surfaces. The importance of edge density fluctuations on the effective control of the current density profile by the rf waves in the core of the plasma has been studied extensively. Important developments are under way to formulate and implement a consistent neoclassical current drive theory for non-Maxwellian plasmas, incorporating nonlocal effects resulting from finite orbit width, including the contribution from multiple species, all while keeping the structure of the highly benchmarked numerical framework already developed in LUKE. Applications include a consistent calculation of the bootstrap current in the presence of rf waves and wave-induced transport of fast particles.

In a more distant future, it is envisioned to use the increased power of parallel computing to solve problems with a higher number of dimensions. The domain of applicability of kinetic codes designed for the current drive problem can be significantly extended, near the magnetic axis, in the magnetic islands, or close to the plasma edge where the collisionality increases. With less restrictive assumptions, the equations will have a simpler form than those used today, but their numerical implementation will certainly be much more challenging. Until then, bounce or orbit averaging is used to eliminate the fast dynamics along magnetic field lines, such that the phase-space dynamics is principally dominated by coulomb collisions, which contribute to regularizing the matrix representing the differential operator. With a higher number of dimensions, the problem might be solved by merging current drive modeling with gyrokinetic calculations for turbulence.

ACKNOWLEDGMENTS

The authors would like to thank J.-F. Artaud for fruitful discussions on integrated tokamak modeling and J. Hillairet for

help regarding LH antennas. This work, supported by the European Communities under the contract of association between EURATOM and CEA, was carried out within the framework of the European Fusion Development Agreement. The views and opinions expressed herein do not necessarily reflect those of the European Commission.

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